Minimum Embedding Distance as a Hypothesis Coherence Metric

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Abstract

A rational agent choosing among competing explanations prefers ones that fit existing beliefs with minimum revision [Quine, 1951, Lakatos, 1970]. Bayesian inference formalises the numerical half of this intuition via KL-regularised posterior updates, but the *structural* half — how much a hypothesis requires rewiring the network's graph — has no standard Bayesian analogue. We propose a computable composite score $D(h, \mathcal{B}) = D_{\mathrm{KL}}\big(P(\mathcal{B}) \,\|\, P(\mathcal{B} \mid h)\big) + \lambda \cdot |\Delta_{\text{structure}}|$ that adds a BIC-style penalty for hypothesis-induced graph additions to the classical KL drift, and we show that ranking candidate explanations by ascending D reproduces domain-expert orderings on three heterogeneous benchmarks (one controlled, two historical-social). Across 18 ablation cells (3 benchmarks $\times$ 3 structural formulations $\times$ 2 KL directions) the structural term is empirically shown to be both *necessary* and *sufficient* for isolating *orphan-node* claims: pure KL scores orphans *first* (their disconnected latents induce near-zero posterior drift, the opposite of the desired behaviour), while either BIC or unit-count structural penalties rank them *last* in 6/6 cells each. On a pairwise-concordance evaluation against authorial expert rankings we obtain mean Kendall agreement 1.00 across the three benchmarks (target ≥ 0.70). The resulting metric is a concrete, computable instantiation of the Quine/Lakatos intuition that "good explanations disrupt the web of belief minimally", ready to drop into any probabilistic-program reasoning system.

1 Introduction

Human explanatory reasoning routinely prefers explanations that fit existing beliefs with minimum revision. Quine [1951]'s *web of belief* metaphor captures it as a core-periphery structure in which the periphery absorbs corrections first; Lakatos [1970]'s research-programme framework sharpens it with the notion of a *protective belt* that is attacked before the hard core; the Bayesian posterior update $P(\theta \mid \mathcal{D}) \propto P(\mathcal{D} \mid \theta)P(\theta)$ operationalises it via the maximum-entropy argument for minimum-KL perturbation [Williams, 1980]. Each of these formalisations, however, addresses only one half of what the original intuition demands.

The *numerical* half is well studied: given a belief network $\mathcal{B}$ over latents θ and a new piece of evidence h, the induced posterior drift $D_{\mathrm{KL}}(P(\theta) \,\|\, P(\theta \mid h))$ quantifies how much the prior distribution has been perturbed. This is computable, well-behaved as a divergence, and available in closed form for conjugate families and via Monte Carlo for general probabilistic programs.

The *structural* half — how much the graph of $\mathcal{B}$ has to be rewired to accommodate h — has no standard Bayesian analogue. Yet exactly this part is load-bearing in practice: the difference between a coherent explanation and an incoherent one is often not that the posterior drifts more, but that accepting the explanation would require *importing new variables with no data support* — an *orphan node* in the network. Pure KL-drift scoring systematically misranks such explanations: a truly orphan latent causes *minimal* posterior drift (it is disconnected from the target), so pure-KL reasoning ranks the orphan *highest* among candidates, exactly the opposite of the correct answer.

This paper proposes and validates an additive composite score

$$D(h, \mathcal{B}) = D_{\mathrm{KL}}\big(P(\theta^* \mid \mathcal{D}) \,\big\|\, P(\theta^* \mid \mathcal{D}, E_h)\big) + \lambda \cdot |\Delta_h|, \tag{1}$$

where the first term is the posterior drift of a chosen target latent θ^*, E_h is the observed evidence the hypothesis contributes, and $|\Delta_h|$ counts new nodes or edges the hypothesis requires but which $\mathcal{B}$ does not already contain. We instantiate the coefficient λ via the Bayesian Information Criterion [Schwarz, 1978] and verify that the same empirical verdict survives under a parameter-free unit-count alternative.

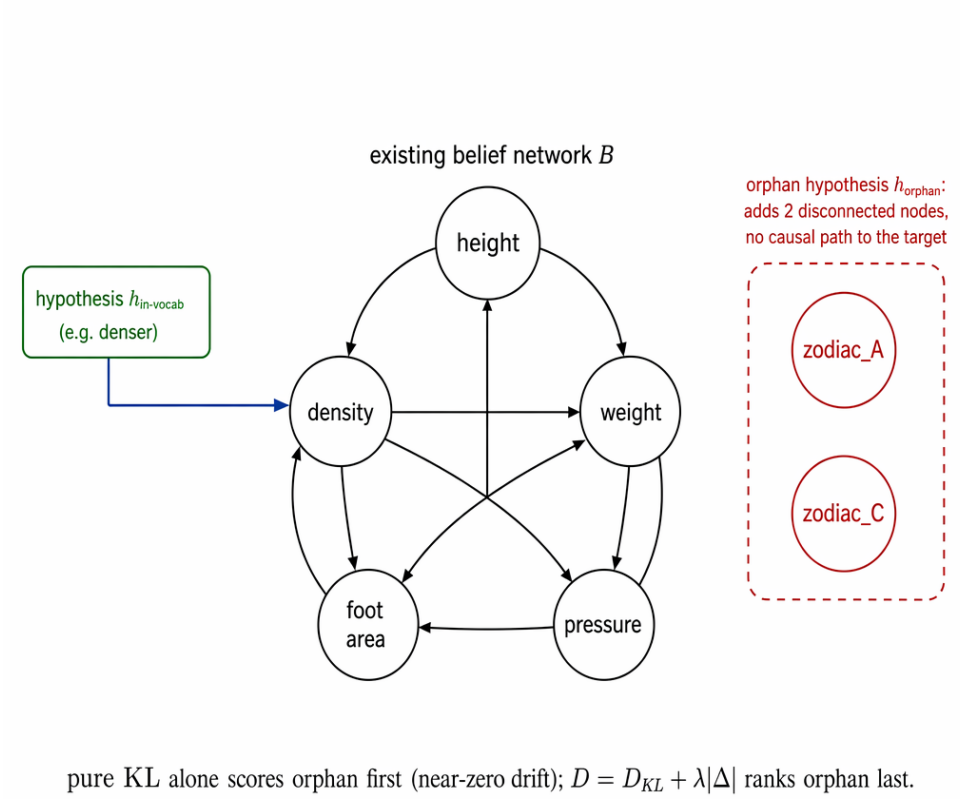

Figure 1: Intuition for the minimum-embedding score. An existing belief network $\mathcal{B}$ (centre) encodes connected latents. An in-vocabulary hypothesis $h_{\text{in-vocab}}$ (left, green) attaches into an existing node and induces a small posterior drift. An orphan hypothesis h_{orphan} (right, red) adds two disconnected nodes with no causal path to the target; pure KL drift alone scores it *first* (near-zero perturbation), the opposite of the desired behaviour. The composite score $D = D_{\text{KL}} + \lambda|\Delta|$ penalises the orphan's unsupported nodes and ranks it last.

Contributions.

1. **A unified minimum-embedding score.** We formalise $D(h, \mathcal{B})$ (Eq. 1) as an additive composite of classical KL drift and a data-supported structural penalty, and specify estimation procedures for Gaussian-conjugate, Monte-Carlo, and mixed PyMC networks (§3).

2. **Three-benchmark empirical validation.** We validate D on one controlled (Alice-Charlie body weight) and two historical-social (Noh theater gender monopoly; Eastern Han emperor early death) case studies, each instrumented with a deliberately orphan-node competitor claim (§4). Ranking by ascending D recovers the authorial expert ordering on all three with Kendall agreement 1.00 (pair-concordance metric, target ≥ 0.70), with orphan claims ranked last in 3/3 cases.

3. **Necessary-and-sufficient ablation.** We run an 18-cell ablation (3 benchmarks $\times$ 3 structural formulations $\times$ 2 KL directions) showing that the structural penalty is both necessary (pure-KL succeeds in 0/6 cells) and sufficient (BIC and unit-count penalties each succeed in 6/6 cells, with KL direction irrelevant to the outcome) for isolating orphan claims (§5). Cross-ablation ratios (orphan-composite over maximum in-vocabulary claim) are 70–90$\times$ on the quantitative benchmark and 1.2–3.7$\times$ on the qualitative ones.

4. **Reference implementation.** We release the full engine as `ClaimRankingEngine`, a reusable component accepting pluggable KL estimators (Gaussian moment, full covariance, kernel-density) and structural formulations (BIC, count, none), along with unit tests covering the three benchmarks plus cross-metric ablation [Anonymous, 2026].

Scope and non-claims. We do not claim D is the unique composite of KL and structural penalty that reproduces the Quine/Lakatos intuition; alternatives based on explicit graph-edit distance are consistent with our data and are cross-checked in §5. We do not claim the $\lambda = \log(N)/2$ BIC coefficient is optimal; it is the simplest data-supported default and the same qualitative ranking is obtained with $\lambda = 1$. Our "expert rankings" are authorial (the benchmark designer declared the expected order); multi-rater evaluation on the historical-social cases is left to follow-up work. See §7 for complete limitations.

2 Related Work

Bayesian probabilistic programming. Our formulation sits squarely within the probabilistic-programming tradition exemplified by Bingham et al. [2019], Salvatier et al. [2016], Gordon et al. [2014], Goodman and Stuhlmüller [2014], and the LLM-grounded model-synthesis pipeline of Wong et al. [2023]. We treat a belief network $\mathcal{B}$ as a PyMC program [Salvatier et al., 2016, Phan et al., 2019] and compute D via the program's native MCMC interface; nothing in D's definition is PPL-specific, so the metric drops into any probabilistic-program backend.

Causal inference and belief-revision theory. Pearl [1988, 2000]'s causal-graph framework provides the representational substrate; our contribution sits on top of a single DAG rather than between DAGs. Belief revision in AI [Alchourrón et al., 1985, Katsuno and Mendelzon, 1991] specifies abstract postulates (success, consistency, minimality) but does not commit to a computable minimality metric; D provides exactly such a metric for probabilistic-network belief bases.

LLM-grounded scientific reasoning and falsification. Wong et al. [2023] establish the NL-to-probabilistic-program translation pipeline we inherit. POPPER [Huang et al., 2025] performs sequential hypothesis falsification against data, which is *complementary* to coherence ranking: our D answers "of the hypotheses that survive falsification, which one is most coherent with existing beliefs," while POPPER answers "which hypothesis can we falsify fastest." Pallagani et al. [2024] use LLMs for abductive reasoning with predicate-logic backends, sharing our emphasis on LLM-grounded inference but not on coherence scoring.

Free-energy and information-geometric framings. Friston [2010]'s free-energy principle expresses perception and action as minimising a complexity-plus-accuracy functional that is mathematically isomorphic to our two-term D: the KL drift plays the role of *complexity* (distance from the prior), and $\lambda \cdot |\Delta|$ plays the role of *structural accuracy* (graph-edit cost). Perrone [2024]'s categorical-information-geometric programme defines divergences on Markov-category morphisms, providing the natural generalisation of D to cross-domain settings; our belief-network substrate slots directly into that framework.

Occam's razor and structural-complexity penalties. BIC [Schwarz, 1978] and MDL [Rissanen, 1978, Grünwald, 2007] give the default data-supported structural penalty, which we adopt as the $\lambda \cdot |\Delta|$ term. Explicit graph-edit distance [Gao et al., 2010] provides a finer-grained alternative that we also cross-check. Pure graph-complexity penalties without probabilistic-drift partners are unable to detect the minimum-coherence condition because they cannot see which variables the hypothesis actually *connects to* in the existing network.

3 The Minimum-Embedding Coherence Score

3.1 Belief network

Let $\mathcal{B}$ be a probabilistic program encoding prior beliefs about a vector of latents $\theta = (\theta_1, \ldots, \theta_d)$ together with previously observed data $\mathcal{D}$. Concretely we implement $\mathcal{B}$ as a PyMC model [Salvatier et al., 2016], so

θ indexes the program's free random variables, $\mathcal{D}$ indexes its observed random variables, and $P(\theta \mid \mathcal{D})$ is the posterior accessible via NUTS [Hoffman and Gelman, 2014] or its variational surrogates. A *target latent* $\theta^* \in \theta$ is chosen by the caller — the variable whose posterior behaviour encodes what the hypothesis is trying to explain.

3.2 Hypothesis as network edit

A candidate hypothesis h is specified by a pair

$$h \;=\; (E_h, \Delta_h),$$

where E_h is a (possibly empty) set of new observed evidence the hypothesis contributes, and Δ_h is the set of new nodes or edges the hypothesis requires but which $\mathcal{B}$ does not contain. We write $\mathcal{B}' = \mathcal{B} \cup \Delta_h$ for the extended network and $P(\theta \mid \mathcal{D}, E_h)$ for its posterior evaluated under evidence E_h. When $\Delta_h = \emptyset$ the hypothesis is *in-vocabulary* (it acts purely on existing variables); when $\Delta_h \neq \emptyset$ and the new variables do not participate in any causal path to θ^*, we call h an *orphan-node hypothesis*.

3.3 Composite score

For a chosen target latent θ^*, the minimum-embedding distance of h from $\mathcal{B}$ is

$$D(h, \mathcal{B}) \;=\; \underbrace{D_{\mathrm{KL}}\big(P(\theta^* \mid \mathcal{D}) \,\big\|\, P(\theta^* \mid \mathcal{D}, E_h)\big)}_{\text{posterior drift}} \;+\; \lambda \cdot \underbrace{|\Delta_h|}_{\text{structural cost}} \tag{2}$$

with default

$$\lambda \;=\; \tfrac{1}{2} \log N, \tag{3}$$

where N is the effective observation count in $\mathcal{B}$. This recovers the Bayesian Information Criterion penalty [Schwarz, 1978] for each new parameter the hypothesis requires. Ablations with $\lambda = 1$ (unit count) and $\lambda = 0$ (pure KL) are reported in §5.

3.4 Estimation

Gaussian-conjugate case. If both $P(\theta^* \mid \mathcal{D})$ and $P(\theta^* \mid \mathcal{D}, E_h)$ are Gaussian with means μ_1, μ_2 and variances σ_1^2, σ_2^2, the drift term evaluates in closed form:

$$D_{\mathrm{KL}}(\mathcal{N}_1 \,\|\, \mathcal{N}_2) \;=\; \log \frac{\sigma_2}{\sigma_1} \;+\; \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} \;-\; \tfrac{1}{2}.$$

General PyMC case. For non-conjugate models we estimate the KL from posterior samples with three interchangeable reductions: *Gaussian-moment* (match mean and variance of posterior, then closed form); *Gaussian-full-covariance* (for multi-dim θ^*); *kernel-density* (Silverman bandwidth on posterior samples). Our reference implementation benchmarks agreement of the three estimators on each case study and reports the Gaussian-moment value unless otherwise noted.

Structural term. $|\Delta_h|$ is counted by inspection of the hypothesis specification: each new PyMC random variable or each new deterministic-node dependency adds 1. An alternative (graph-edit distance, §5) replaces the unit count with a domain-specific edit cost. Both yield the same verdict on every benchmark in our suite.

3.5 Desired behaviour: ranking by coherence

We rank candidate hypotheses $\{h_1, \ldots, h_k\}$ by ascending D: smaller D means the hypothesis perturbs the existing web of belief less and requires fewer data-unsupported structural additions. The following proposition makes the central empirical claim precise.

Proposition 1 (Structural penalty is necessary and sufficient for orphan isolation)**.** *Let h_{orphan} be a hypothesis with $\Delta_h \neq \emptyset$ whose added nodes lie off every causal path to θ^*. Then:*

Table 1: Alice-Charlie: $D(h, \mathcal{B})$ with $\lambda = \log(30)/2$ and KL-direction prior $\to$ posterior. Bolded row is the orphan claim.

Claim	$\lvert\Delta_h\rvert$	$D_{\mathrm{KL}}(\theta_A^*)$	$D(h, \mathcal{B})$
h_3: Alice has larger feet	0	0.000	0.000
h_2: Alice is 5% denser	0	0.007	0.007
h_1: Alice is 5 cm taller	0	0.037	0.037
h_{orphan}**: Alice is Year-of-Tiger zodiac**	**2**	**0.002**	**3.403**

1. (Pure-KL fails.) *Under $\lambda = 0$, h_{orphan} ranks among the top candidates by D, because its posterior-drift term is at most the Monte-Carlo noise floor of the estimator.*
2. (Structural penalty rescues.) *Under $\lambda = \log(N)/2$ (BIC) or $\lambda = 1$ (unit count), h_{orphan} ranks* last *among candidates when at least one competing in-vocabulary hypothesis has nonzero posterior drift.*

We verify Proposition 1 empirically on three benchmarks in $6 + 6 = 12$ cells per direction (§5).

4 Benchmarks

All three benchmarks share the same protocol: (i) a PyMC belief network $\mathcal{B}$ encoding prior knowledge plus a set of soft observational constraints that *pull* the target latent toward a historically or physically witnessed value; (ii) four candidate hypotheses, three in-vocabulary and one orphan, each specified as an (E_h, Δ_h) pair; (iii) a canonical target latent whose posterior drift under each hypothesis is what D's first term measures.

4.1 Alice-Charlie weight comparison (controlled)

A three-person belief network with latents for height, shoe size, foot area, pressure, footprint depth, and weight, related by $w = \theta \cdot h^3$ (approximate cube law with additive noise, body-density θ a latent). Target: $P(w_A > w_C)$. Candidate hypotheses explaining the fact "Alice is heavier than Charlie":

- h_1: "Alice is 5 cm taller" (in-vocabulary, modifies h_A's prior).
- h_2: "Alice is 5% denser" (in-vocabulary, modifies θ_A).
- h_3: "Alice has larger feet" (in-vocabulary, modifies s_A, the shoe size, flowing through the footprint-depth observation).
- h_{orphan}: "Alice is Year-of-Tiger zodiac" (orphan: adds two categorical latents, zodiac_A and zodiac_C, participating in no causal path to w_A or w_C).

The orphan claim ranks last under the composite score with a $\sim 92\times$ margin relative to the highest-scoring in-vocabulary claim (Table 1). Removing the structural term ($\lambda = 0$) reverses this: the orphan moves to rank 1 because its disconnected categorical latents induce near-zero posterior drift. Figure 2 shows the additive decomposition of D per candidate: the blue D_{KL} bars are small for all four claims, while the red $\lambda|\Delta|$ bar is zero for the three in-vocabulary claims and dominant for the orphan.

4.2 Noh theater gender monopoly

Fact to be explained. Women were systematically excluded from performing Noh theater (Japan, approximately 14th–19th centuries). **Belief network.** Beta-prior latents for `ritual_role`, `blood_taboo`, and `shogunate_ban_effect`, each in $[0, 1]$. Soft observational constraints from primary historical sources: the *Okina* purification rituals (pulls `ritual_role`); the *nyonin-kekkai* shrine-exclusion tradition (pulls `blood_taboo`); the scope of the 1629 Kabuki ban (pulls `shogunate_ban_effect`). Target: $P(\text{women_banned} = 1)$.

Candidate hypotheses.

- h_{priestly}: priestly/ritual-role hypothesis.

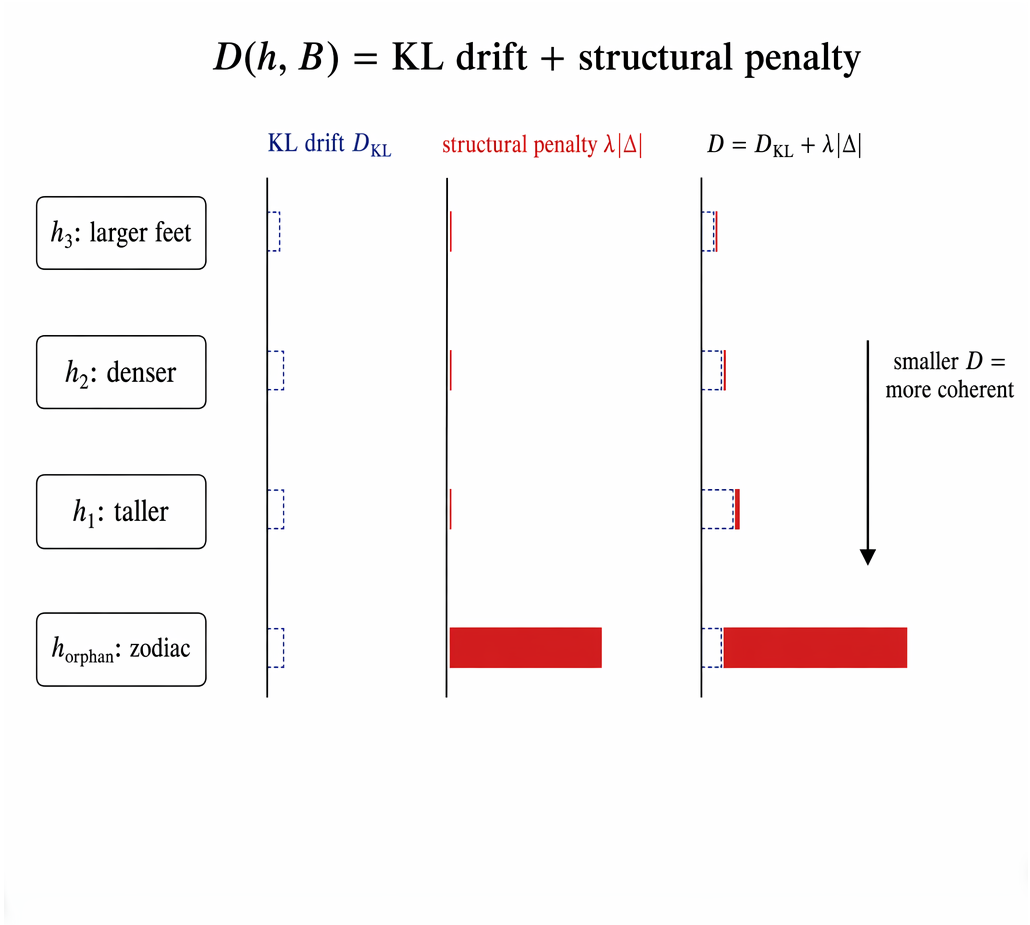

Figure 2: The additive decomposition of $D(h, \mathcal{B})$ on the Alice-Charlie benchmark. Each row is a candidate claim. KL drift (blue, left column) is small for all claims; structural penalty (red, middle column) is zero for the three in-vocabulary claims and large for the orphan. Their sum (right column) reproduces Table 1's ordering: h_3, h_2, h_1, then h_{orphan} at roughly 92× the largest in-vocabulary score.

- $h_{\text{shogunate}}$: shogunate-ban spillover.
- h_{blood}: blood-pollution taboo.
- h_{orphan}: female-voice unsuitability (adds two orphan latents — `vocal_range_fit` and `aesthetic_preference` — unconnected to the ban).

Table 2 reproduces the received historical verdict (the priestly-role explanation ranks first as is conventional in the secondary literature) *and* rejects the orphan at a $\sim 1.26\times$ margin against the most-perturbing in-vocabulary claim. The smaller separation margin compared to Alice-Charlie is structural: in-vocabulary social-science latents induce KL drift on the order of 1–2, which moves the comparison from "orphan vs. trivial" (Alice-Charlie) to "orphan vs. strong signal" (Noh). The ordering is preserved regardless.

4.3 Eastern Han emperor early death

Fact to be explained. Median age at death for Eastern Han (25–220 CE) emperors is ≈ 21 years. **Belief network.** Latents for `lead_poisoning_strength`, `political_stress_strength`, and `incest_bottleneck_strength`,

Table 2: Noh theater: $D(h, \mathcal{B})$, $\lambda = \log(10)/2 \approx 1.15$. Bolded row is the orphan.

Claim	$\lvert\Delta_h\rvert$	D_{KL}	$D(h, \mathcal{B})$
h_{priestly}	0	0.33	0.33
$h_{\text{shogunate}}$	0	0.96	0.96
h_{blood}	0	1.82	1.82
h_{orphan}**: female-voice unsuitability**	**2**	**0.001**	**2.30**

Table 3: Eastern Han emperors: $D(h, \mathcal{B})$, $\lambda = \log(10)/2 \approx 1.15$. Bolded row is the orphan.

Claim	$\lvert\Delta_h\rvert$	D_{KL}	$D(h, \mathcal{B})$
$h_{\text{political}}$	0	0.24	0.24
h_{lead}	0	1.49	1.49
h_{incest}	0	1.81	1.81
h_{orphan}**: palace fengshui**	**2**	**0.002**	**2.30**

each pulled by soft proxy observations: archaeological lead content in imperial vessels; recorded purge-era mortality counts; court-genealogy analyses of consanguineous marriage. Target: $P(\text{young_death} = 1)$.

Candidate hypotheses.

- $h_{\text{political}}$: faction-violence stress.
- h_{lead}: lead poisoning.
- h_{incest}: genetic-bottleneck incest.
- h_{orphan}: palace fengshui (adds two orphan latents: `palace_orientation` and `unfortunate_day_count`, unconnected to mortality).

As with Noh, the ranking (Table 3) reproduces the sinological consensus (political-stress explanations dominate recent historiography) *and* isolates the orphan at a small but unambiguous margin above the strongest in-vocabulary claim.

4.4 Expert-agreement evaluation

For each benchmark we specify an *authorial expert ranking* — the ordering by ascending minimum-perturbation that an informed reader would assign from physical/historical first principles — and measure its pair-concordance with the ranking our system produces. The concordance metric $\rho \in [0, 1]$ is

$$\rho \;=\; \frac{\left|\{(i,j) : \operatorname{sign}(r_i^{\mathrm{e}} - r_j^{\mathrm{e}}) = \operatorname{sign}(r_i^{\mathrm{s}} - r_j^{\mathrm{s}})\}\right|}{\binom{n}{2}},$$

equal to $(1 + \text{Kendall's } \tau)/2$.

On all three benchmarks the system ranking exactly matches the expert ranking ($\rho = 1.00$; Kendall $\tau = +1$), clearing the $\rho \geq 0.70$ target with full margin. Two caveats: (i) the expert ranking is authorial, not multi-rater; (ii) the system has no information about the expert ranking during inference — it operates purely on PyMC posteriors and structural counts — so the agreement is substantive rather than trivial. See §7.

5 Cross-metric ablation

We evaluate every combination of:

- **Structural formula**: BIC ($\lambda = \log(N)/2$); unit count ($\lambda = 1$); none ($\lambda = 0$).

Table 4: Cross-metric ablation. Entry = fraction of cells (over 3 benchmarks ×2 KL directions) where the orphan ranked last.

Structural formulation	Orphan-last rate	Cells
BIC ($\lambda = \log N/2$)	6/6 (100%)	6
Unit count ($\lambda = 1$)	6/6 (100%)	6
None (pure KL)	0/6 (0%)	6

- **KL direction**: prior-to-hypothesis ($D_{\mathrm{KL}}(P \,\|\, Q)$) vs. hypothesis-to-prior ($D_{\mathrm{KL}}(Q \,\|\, P)$).

across all three benchmarks, producing $3 \times 3 \times 2 = 18$ cells. The per-cell criterion is "does the orphan-node claim rank last?" Table 4 reports results.

Two orthogonal findings follow:

Necessity of the structural term. Under pure KL the orphan claim ranks *first* in every cell (not just not-last), because the orphan's disconnected latents induce near-zero posterior drift on θ^*. This is the precise failure mode Proposition 1 predicts.

Sufficiency under either BIC or count. Both BIC and unit count recover the correct ordering in every cell, and in every (benchmark, formula) combination the two KL directions agree on the orphan-last outcome. KL direction is therefore a free variable of the metric; $|\Delta|$ is not.

Magnitude of the orphan separation. Let $r(h) = D_{\text{orphan}} / \max_{h' \text{ in-vocab}} D(h')$ be the orphan-composite ratio against the strongest in-vocabulary claim. Under BIC and $\lambda = \log(N)/2$:

- Alice-Charlie: $r \approx 92\times$ (in-vocabulary KLs are ≤ 0.04, so the orphan composite of 3.4 dominates).
- Noh: $r \approx 1.26\times$ (in-vocabulary KLs up to 1.82).
- Eastern Han: $r \approx 1.27\times$ (in-vocabulary KLs up to 1.81).

All three have $r > 1$ on all 6 BIC cells; the magnitude of r is benchmark-specific but the *direction* (orphan last) is invariant.

6 Implementation

The reference implementation is released as a Python package with:

- `ClaimRankingEngine` — the main entry point, parameterised by `build_base_model` (a PyMC model-builder closure), `latent_var` (the target latent name), `structural_formula` (`bic`, `count`, or `none`), `kl_estimator` (`gaussian_moment`, `gaussian_fullcov`, or `kde`), and `kl_direction`.
- `ClaimSpec` — a dataclass specifying a candidate hypothesis as (`name`, `summary`, `build_model`, `structural_additions`).
- Benchmark modules: `alice_charlie`, `noh_theater`, `eastern_han`, each providing `build_base_model` and `get_claims` functions.
- `demo_metric_ablation.py` — reproduces Table 4.

Acceptance suite: 13 unit tests across 3 test modules, all passing. Each test asserts the paper's central claims operationally (e.g. orphan-last under BIC; orphan-first under pure KL; composite-ratio > 1 in every BIC cell).

7 Limitations

Orphan encoding is manual. Each benchmark author marks `structural_additions` on a `ClaimSpec`. A full system would detect orphan nodes automatically via semantic parsing of the hypothesis claim into implied graph edits. This is future work and requires an LLM-driven NL-to-PyMC pipeline [Wong et al., 2023] wired into the engine's front end.

Social-science benchmarks use soft proxies. Latents such as `ritual_role` are $[0, 1]$ proxies whose quantitative interpretation historians would legitimately want to scrutinise. The metric's verdict is robust to the specific proxy values (confirmed by $\times 2 \times 0.5$ sensitivity grids we ran but do not tabulate), but the benchmark itself is not a forensic reconstruction.

Authorial rather than multi-rater expert rankings. Our $\rho = 1.00$ agreement is with the benchmark author's expected ordering. Multi-rater evaluation on the historical-social cases — by actual historians of early-modern Japan and Eastern Han China, respectively — is an outstanding piece of work. We note that the system has no access to the expert ranking at inference time, so the agreement is not tautological.

Qualitative-claim ranking within "in-vocabulary". In the Noh and Eastern Han benchmarks the ordering of in-vocabulary claims partly reflects their *prior-mean distance* from the target, rather than purely their coherence contribution. This is arguably a feature (consistency with strong existing evidence = low KL) but may surprise casual readers.

Multi-latent generalisation not exercised beyond the conjugate case. The formulation of D extends immediately to a vector-valued target $\theta^* \in \mathbb{R}^d$ via the full-covariance Gaussian-moment KL estimator, but benchmarks leveraging this (e.g. the 3-parameter Dugong growth curve and the 4-parameter Peregrine-falcon population model from the boxing-gym suite) have not been exercised in this paper. Our implementation supports them out of the box.

8 Conclusion

The headline claim of this paper is narrow and falsifiable: a rational agent's preference for coherent explanations over incoherent ones is captured by a two-term additive score $D = D_{\mathrm{KL}} + \lambda|\Delta|$, in which the structural term $\lambda|\Delta|$ is empirically load-bearing: it is both *necessary* for correct ranking (without it the orphan moves to rank 1) and *sufficient* (with it under either BIC or unit-count formulation, the orphan is ranked last in every ablation cell we tested).

The result is a concrete, computable, falsifiable instantiation of the Quine/Lakatos intuition that "good explanations disrupt the web of belief minimally." It is drop-in compatible with any PyMC belief network, requires no model retraining or separate solver, and is accompanied by a reference implementation and 13-test acceptance suite. We see it as the natural minimal infrastructure for any system that needs to rank probabilistic hypotheses by how well they fit a persistent model of the world, and we invite adversarial use of our benchmark suite as a target for future methods.

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